

KARAN SHAH

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Technical Skills

Languages: Java, C++, JavaScript, SQL

Frameworks & Systems: Spring Boot, Hibernate/JPA, React.js, Apache Kafka, Microservices

Databases & Tools: MySQL, PostgreSQL, MS SQL Server, Docker, Git, Postman, Maven

Education

Nirma University

Jun 2020 – Jun 2024

B.Tech in Computer Science and Engineering

7.72 CGPA

Experience

Tata Consultancy Services (TCS)

Nov 2024 – Present

Systems Engineer (Backend Developer)

Gandhinagar, India

- Designed and implemented a modular backend for a **Complaint Management System**, utilizing **Spring Boot** for service orchestration and **Spring Security** for robust Role-Based Access Control
- Integrated **Apache Kafka** to decouple microservices, facilitating asynchronous event processing and ensuring zero data loss during high-traffic bursts.
- Optimized **Hibernate/JPA** database interactions, automating data entry pipelines which reduced query latency and cut manual operational workload by **30%**.
- Designed REST APIs and managed schema migrations for complex datasets across **MySQL** and **PostgreSQL** environments.

Kavayahcloud PVT LTD.

Jan 2024 – Oct 2024

Software Engineer Intern

Gandhinagar, India

- Developed a full-stack maintenance scheduling module using **JavaScript** and **MS SQL Server**, enabling administrators to automate patch deployment planning.
- Integrated **Power BI** analytics into the core web application via **OAuth2** authentication, providing secure, real-time visualization dashboards for stakeholders.

Civic Infotech

June 2023 – July 2023

Software Engineer Intern

Ahmedabad, India

- Developed a high-performance Single Page Application (SPA) for live auction bidding using **React.js** and Tailwind CSS.
- Implemented real-time data visualization using **Chart.js** to render live bid fluctuations and auction insights.

Projects

GeoFleet – Real-Time Location Platform | *Java 21, Spring Boot, Kafka, gRPC*

Distributed Systems

- Architected an event-driven microservices ecosystem for a ride-sharing simulation, utilizing Apache Kafka to decouple high-velocity ingress traffic from complex matching logic, ensuring system resilience during simulated traffic bursts.
- Engineered a high-throughput Driver Ingestion Service using gRPC bidirectional streaming instead of REST; reduced network payload size by 40% via Protocol Buffers and enabled persistent connections for continuous location telemetry.
- Orchestrated a real-time feedback loop using WebSockets (STOMP protocol) to push matching events to clients, eliminating the overhead of HTTP short-polling and reducing redundant server load.

Fashion semantic search | **Video** | *Python, Flask, BERT, Web Scraping*

ML & NLP

- Built a scalable web scraping pipeline to extract structured JSON data from **10,000+** fashion items across multiple e-commerce platforms.
- Integrated a **BERT** Large Language Model (LLM) to power semantic search capabilities, allowing users to query items based on context rather than just keyword matching.

MirrorLink – Real-Time Screen Sharing | *WebRTC, React, Node.js, Socket.IO*

Web Applications

- Built a real-time screen sharing web application using **WebRTC** to enable direct, high-speed video streaming between devices.
- Created a **Node.js** and **Socket.IO** backend server to manage secure room codes and connect pairing devices.
- Developed a responsive frontend in **React** that utilizes browser APIs (`navigator.mediaDevices.getDisplayMedia`) to capture and display live screen feeds.

- Developed a ready-to-deploy broker integration module for the **Zerodha Kite Connect API**, building REST clients and WebSocket tick listeners to run inside a simulated paper-trading sandbox.
- Implemented a breakout momentum algorithm utilizing **Opening Range Breakout (ORB)**, **VWAP**, and **ATR** indicators to identify high-probability entries using simulated market feeds.
- Designed a robust risk management layer featuring dynamic position sizing based on a fixed capital risk threshold, trailing exit stops, and a portfolio-wide safety kill-switch.
- Built a backtesting engine to evaluate strategy performance over a 5-year historical window, generating key trade telemetry (slippage, win rate, drawdown) logged automatically to **SQLite** and **CSV**.

- Engineered a robust browser automation system using **Playwright** to search job portals, scrape listings, and navigate multi-page results with resilient error-handling for slow-loading pages.
- Optimized scanning speed by executing custom **JavaScript** checks on the search page to detect "Applied" or "Saved" badges, skipping redundant listings without opening them.
- Integrated a **Large Language Model (LLM)** API to parse job descriptions and evaluate them against a target resume, calculating a relevance score from 0% to 100% based on technical fit.
- Designed a conditional workflow that triggers the application process only for listings exceeding a 60% match threshold, preserving daily application limits for high-match opportunities.

- Designed chat analyser using Python to analyse chat based on time and person
- Implemented sentiment analysis using NLTK library of NLP for finding different users sentiment based on chats

Coding Profiles

- **Codeforces**: Max Rating 1438 (Specialist)
- **LeetCode**: Max Rating 1692 (Top 15%)

Achievements

- **LeetCode**: Max Rating **1692** (Top 15% globally). Ranked **1,541** out of 21,000+ in Weekly Contest.
- **Codeforces**: Max Rating **1438** (Specialist). Ranked **1,394** out of 30,000+ in Div 2 Contest.
- **Facebook Hacker Cup**: Qualified for the 2022 edition among global participants.
- **Problem Solving**: Solved **1000+** algorithmic problems across various coding platforms.
- **Awards**: Top 10 Finisher in "Code Uncode 2024" organized by TLE Eliminators.